MLOPS ROADMAP

🧠 MACHINE LEARNING ENGINEER INTERVIEW CHEATSHEET (2025 Edition)

📌 1. Foundations of ML (Refresh Quickly)

Topic	What You Must Know
Types of Learning	Supervised, Unsupervised, Semi-supervised, Reinforcement
Common Tasks	Classification, Regression, Clustering, Ranking

Data Preprocessing	Imputation, Scaling (`StandardScaler` , `MinMaxScaler`) Encoding
Splitting Data	`train_test_split` , Stratified K-Fold, TimeSeriesSplit
Model Validation	Cross-validation, Grid/Random/Optuna Search
Bias-Variance Tradeoff	Underfit = High Bias; Overfit = High Variance

🧠 2. Must-Know ML Algorithms (With Use Cases)

Category	Algorithms	Use Case (2025-relevant)
Linear	Linear/Logistic Regression	Simple tabular datasets
Tree-based	Random Forest, XGBoost, LightGBM	Tabular, high-performance models
Distance	KNN, DBSCAN	Recommenders, anomaly detection
Dim. Red.	PCA, t-SNE, UMAP	Preprocessing + visualization
Ensemble	Bagging, Boosting, Stacking	Kaggle/tabular problems
Deep Learning	CNN, RNN, Transformers	NLP, Vision, Time Series

🆕 2025 Trend: **Transformer-based tabular models** (TabNet, FTTransformer) are gaining adoption in industry.

📊 3. Evaluation Metrics Cheat Table

Problem	Metrics to Know
Classification	Accuracy, Precision, Recall, F1, ROC-AUC, PR-AUC
Regression	RMSE, MAE, R^2, MAPE
Imbalanced Data	F1, ROC-AUC, PR-AUC

🆕 2025 Update: PR-AUC is often **preferred over ROC-AUC** in imbalanced data scenarios.

🛠 4. Tooling & Libraries (Must-Have)

Task	Tools to Know (2025)
Model Building	Scikit-learn, XGBoost, LightGBM, CatBoost
Deep Learning	PyTorch 🔥, TensorFlow/Keras
Experiment Tracking	MLflow, Weights & Biases
Model Deployment	FastAPI, Docker, TorchServe, BentoML
Feature Store	Feast, Tecton

🧩 6. ML System Design – 2025 Hot Topics

✅ Be ready for **open-ended ML design questions** like:

🔧 Sample System Design Questions:

- How would you build a **real-time fraud detection system**?
- Design a **machine learning pipeline** for churn prediction.
- How would you handle **continuous learning / model drift**?
- Build a **recommendation engine** for a music app.

Must Discuss:

- Data ingestion (streaming or batch)
- Offline vs Online models
- Feature engineering & storage
- Deployment: REST API, latency, scalability
- Model retraining & monitoring
- Experiment tracking

🔍 7. Behavioral Interview (STAR Format)

Question	Tip
"Tell me about a time you failed."	Show learning & recovery.
"Describe an ML project end-to-end."	Structure: problem → data → model → results → impact.
"How do you work with cross-functional teams?"	Mention communication + business alignment.

🆕 **2025 Bonus**: Be prepared to talk about **AI ethics, model bias**, and **responsible AI**.